The Intangible Gap

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Motivation

- Corporate investment in the U.S. is shifting toward intangible capital (Corrado and Hulten, 2010; Falato et al., 2022; Crouzet and Eberly, 2019)
- Intangible investment is known to be **risky**
  - Patent-value dispersion rose $\sim$70% from 1975 to its early-2000s peak
  - Innovation pipelines (e.g., LLMs, RL training) generate highly skewed outcomes
- **This paper**: who invests in intangibles, and why?
  - Cross-sectionally: small firms invest *disproportionately* more in intangibles—this gap has nearly tripled since 2000
  - Mechanism: the real-option value of uncertain intangible *quality* + financing channel produces a size-dependent intangible gap

Intangible Gap

Intangible gap $\equiv$ difference in intangible-to-physical capital ratio (H/K) between small and large firms.

1. **The intangible gap is large and rising.**
   - Median H/K of smallest firms grew **8**$\times$ since 1980, vs. only **2.5**$\times$ for largest firms
   - Median intangible gap: 0.54 pre-2000 $\rightarrow$ 1.43 post-2000
2. **Intangible investment has become increasingly risky.**
   - Firm-level *hkratio* volatility rose from 0.15 (1980) to 0.3 (2024) for small firms; stable for large firms
   - Cross-sectional dispersion in patent values rose $\sim$70% from 1975 to early-2000s peak

This Paper

Theory: a firm-dynamics model with two capital types and a novel intangible-quality shock.

Two channels jointly produce the intangible gap:

(i) **Real option**: intangible-quality shock ω + exit option $\Rightarrow$ convex payoff to intangible investment. Especially valuable for small firms near the exit margin.

(ii) **Financing**: Intangible capital can be financed through the internal capital market, while physical capital still relies on external financing collateralized by tangible assets.

Evidence: uncertainty raises *intangible* investment for firms near exit.

Quantitatively: rising riskiness of intangible capital alone explains 60% of the post-2000 rise in the intangible gap.

Related Literature

- **Firm dynamics with financial frictions**: Hopenhayn (1992), Cooley and Quadrini (2001), Gomes (2001), Albuquerque and Hopenhayn (2004), Rampini and Viswanathan (2013)
  - size-dependent heterogeneity in H/K
- **Intangible capital**: Eisfeldt and Papanikolaou (2013), Ai, Croce, and Li (2013), Gourio and Rudanko (2014), Sun and Xiaolan (2019), Falato et al. (2022), Crouzet and Eberly (2023), Eisfeldt, Falato, and Xiaolan (2023)
  - new evidence: rising cross-sectional dispersion in intangible intensity
- **Uncertainty-investment / sign-flipping channels**: Bernanke (1983), Pindyck (1991), Caballero and Pindyck (1996), Leahy and Whited (1996), Bloom (2009); Eisdorfer (2008), Ferreira, Manso, and Silva (2014)
  - *positive* uncertainty-investment relationship for intangibles and a *negative* one for physical capital *within the same firm* simultaneously
- **Corporate borrowing constraints**: Kiyotaki and Moore (1997), Bernanke, Gertler, and Gilchrist (1999), Sun and Xiaolan (2019), Lian and Ma (2021)
  - internal financing channel for intangibles

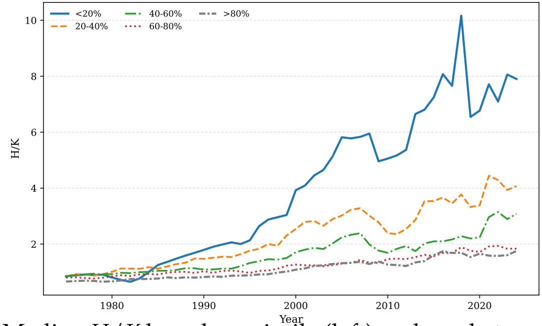
Stylized Facts

Data

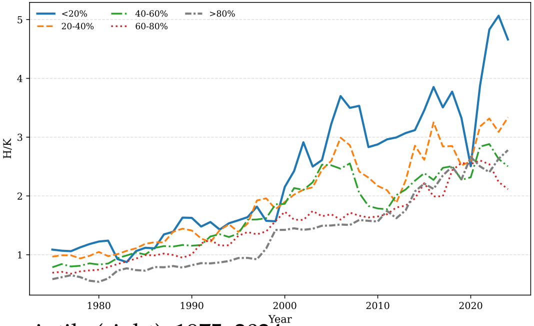
- **Sample**: Compustat Fundamentals Annual, 1975–2024
  - Exclude financials (SIC 6000–6999), utilities (4900–4999), public service (9000+)
- **Intangible capital** Peters and Taylor (2017): $H_{it} = (1 - \delta_h)H_{i,t-1} + I_{H,it}$ including R&D-based knowledge capital + SG&A-based organizational capital + purchased intangibles
  - Alternatives: patent value (Kogan et al., 2017) (KPSS, CPC-subclass $\times$ year normalized); equity-based compensation (STKCO) (Eisfeldt, Falato, and Xiaolan, 2023); balance sheet intangibles (INTAN)
- **Physical capital**: PPEGT

Fact 1: Intangible Gap by Firm Size

(a) By sales percentile



(b) By market-equity percentile



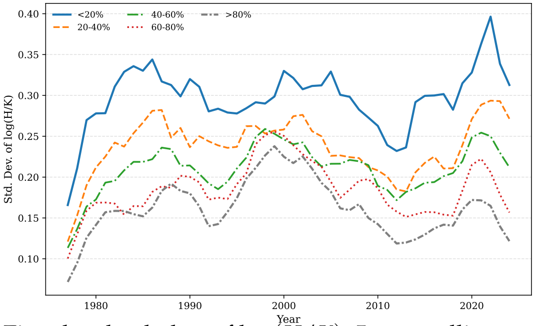
Median H/K by sales quintile (left) and market-equity quintile (right), 1975–2024.

Robust to firm FE, industry composition, life-cycle effects, and within-IPO-cohort comparisons.

Investment rates | Within-firm

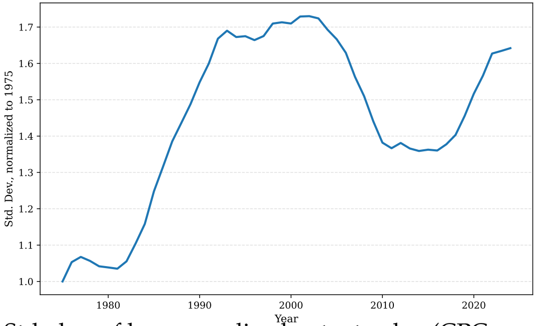
Fact 2: Intangible Investment Is Becoming Riskier

(a) H/K volatility



Firm-level std. dev. of $\log(H/K)$, 5-year rolling window.

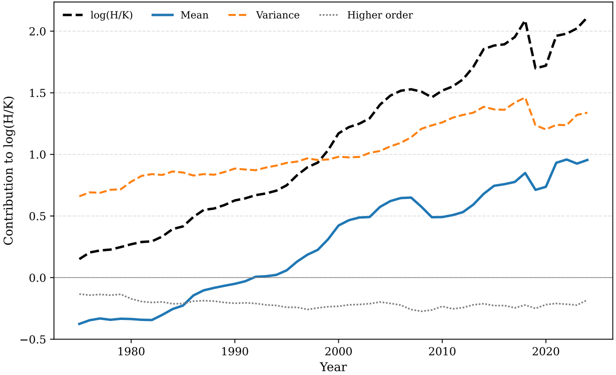
(b) Patent-value dispersion



Std. dev. of log normalized patent value (CPC subclass $\times$ grant-year median), 4-year window; 1975 = 1. Data: Kogan et al. (2017).

Cumulant Decomposition: Variance, Not Just the Mean

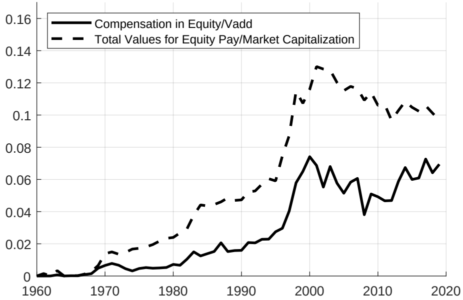
$$\log \overline{H/K}_t = \underbrace{\kappa_1(t)}_{\text{Mean}} + \underbrace{\frac{\kappa_2(t)}{2}}_{\text{Variance}} + \underbrace{\frac{\kappa_3(t)}{6}}_{\text{Skewness}} + \cdots$$



Variance term ($\kappa_2/2$) now rivals the mean in driving aggregate H/K.

Fact 3: Financing Intangibles—Internal Credit Market

- Aggregate share of corporate equity allocated to employee stock-based compensation has **roughly doubled** since 1980 (Eisfeldt, Falato, and Xiaolan, 2023)
- Strongly size-dependent: stkco/AT is larger for smallest firms than for largest
  Summary statistics
- Size gradient closely mirrors the intangible gap



⇒ Motivates the assumption “intangibles can be financed internally” in the model.

The Model

- Two-Period Model
- Dynamic Model

Two-Period Model without Financial Constraint

Setup. Firm enters with h_t, chooses h_{t+1}

Quality shock. ω_{t+1} ($\log \omega \sim N(\mu, \sigma^2)$, $\mu = -\frac{1}{2}\alpha\sigma^2$). Firm operates only if value ≥ 0:

$$V_t = \max_{h_{t+1}} E_t\Big[\max\{(\omega_{t+1}h_{t+1})^\alpha - K(h_{t+1}),\ 0\}\Big], \qquad K(h_{t+1}) = (h_{t+1} - h_t) + \xi$$

BSM form: $V_t = \max_{h_{t+1}} h_{t+1}^\alpha N(d_1) - K(h_{t+1})N(d_2)$.

- **Abandon option:** A call-option-like payoff structure: the firm captures the upside when ω_{t+1} is high, while the exit option truncates the downside when ω_{t+1} is low.

Proposition (Volatility raises investment)

h_{t+1}^* is strictly increasing in σ.

Proposition (Size shrinks the option)

The exit option value $V_t - V_{0,t}$ is strictly decreasing in h_t.

Why an Asymmetric Mechanism?

- **Standard Abel–Dixit–Pindyck real-options:** uncertainty $\uparrow \Rightarrow$ investment $\downarrow$ (irreversibility, waiting option).
- **Our mechanism:** multiplicative *intangible-quality* shock + exit option $\Rightarrow$ convex payoff *for intangibles only*.
  - Physical investment $\Rightarrow$ standard real-options logic ($\downarrow$ with σ)
  - Intangible investment $\Rightarrow$ option payoff ($\uparrow$ with σ, especially for small firms)

Net effect: small firms tilt toward intangibles when σ_ω rises; large firms unaffected.

Dynamic Model: Setup

- **Production:** $F(s_t, k_t, h_t) = s_t k_t^{\alpha_k} h_t^{\alpha_h}$, s_t total productivity shock $\sim$ AR(1).
- **Capital laws of motion:** standard with quadratic adjustment costs ρ_k, ρ_h.
- **Multiplicative intangible quality shock:** $\log \omega_t \sim N(-\sigma_\omega^2/2, \sigma_\omega^2)$, $\mathbf{E}[\omega] = 1$

$$h_{t+1} = \omega_{t+1} \tilde{\boldsymbol{h}}_{\boldsymbol{t+1}}, \qquad \tilde{\boldsymbol{h}}_{\boldsymbol{t+1}} = (1 - \delta_h)\boldsymbol{h_t} + i_{h,t}$$

- **Budget:** $d_t = F(s_t, k_t, h_t) - i_{kt} - i_{ht} - g_h(\cdot) - g_k(\cdot) - \xi$.
- **Financing constraint:** $d_t \geq -\theta h_t$
  - $\theta > 0$ measures the effective borrowing capacity generated by intangible capital (equity-based comp, patent-backed lending, VC)
- **Firm value and exit:** the firm exits when its continuation value is negative,

$$V(s_t, k_t, h_t) = \max_{d_t, i_{k,t}, i_{h,t}} \left\{ d_t + \beta E_t \left[\max\left\{ V(s_{t+1}, k_{t+1}, \omega_{t+1}\tilde{h}_{t+1}), 0 \right\} \right] \right\}.$$

Microfoundation Within-period timing

Investment Wedge: Real Option + Financing

Euler equations (conditional on survival $\mathbf{1}\{\omega_{t+1} > \omega^*\}$):

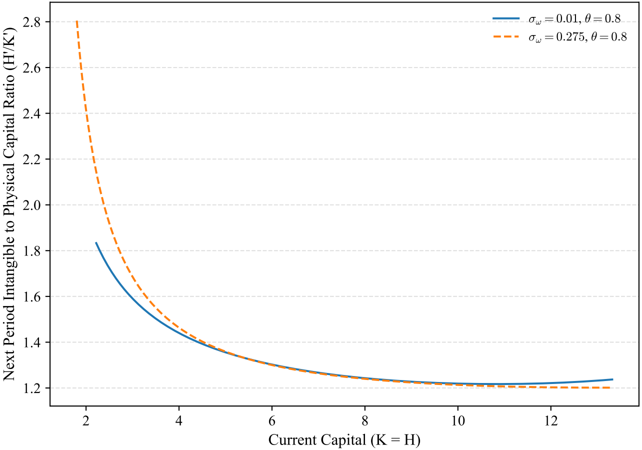
$$(1+\lambda_t)q_{k,t} = \beta E_t\big[\mathrm{MPK}_{t+1} \cdot \mathbf{1}\{\omega > \omega^*\}\big]$$
$$(1+\lambda_t)q_{h,t} = \beta E_t\big[\boldsymbol{\omega_{t+1}}\big(\mathrm{MPH}_{t+1} + \lambda_{t+1}\theta\big) \cdot \mathbf{1}\{\omega > \omega^*\}\big]$$

Two asymmetries between h and k:

- Multiplicative ω_{t+1} on the intangible side $\Rightarrow$ convex payoff in ω
- Additional $\lambda_{t+1}\theta$ term $\Rightarrow$ intangible capital relaxes the constraint

$\Rightarrow$ Small, financially-constrained firms tilt strongly toward intangibles.

Investment Wedge: Real Option + Financing



H'/K' along $K = H$; benchmark financing capacity ($\theta = 0.8$).

No financial frictions | Severe frictions ($\theta = 0.01$)

Empirical Evidence on the Mechanism

Testing the Mechanism: Uncertainty and the Exit Margin

$$I^H_{i,t}/A_{i,t} = \beta\,(U_{i,t} \times \text{Exit}_{i,t}) + \gamma\,U_{i,t} + \lambda\,\text{Exit}_{i,t} + \alpha_i + \delta_t + \varepsilon_{i,t}$$

- **Uncertainty measure:** firms' patent-value dispersion (Kogan et al., 2017); realized return volatility (Alfaro, Bloom, and Lin, 2024).
- **Proximity to exit:** market equity (smaller = closer to exit); a score based on predetermined fundamentals.
- **Result:** the closer a firm is to the exit margin, the more strongly uncertainty raises *intangible* investment; the physical-investment response is insignificant or negative.

Measurement details Summary estimates Full estimates

Calibration and Quantitative Analysis

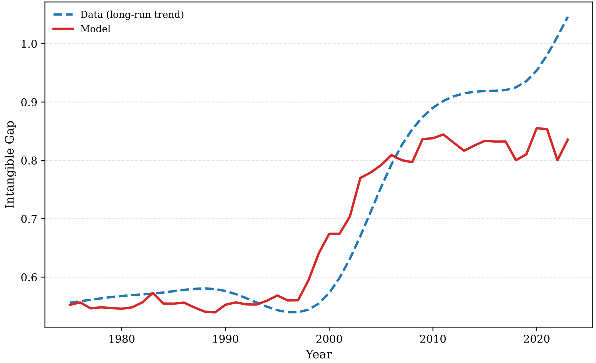
Calibration: One Moment per Parameter (1975–2000)

Parameter	Symbol	Value	Anchored to
Discount factor	β	0.96	4% annual rate
Physical capital share	α_k	0.30	returns to scale (0.71)
Intangible capital share	α_h	0.414	median H/K level (1.10)
Depreciation of physical capital	δ_k	0.12	investment rate of K (0.130)
Depreciation of intangible capital	δ_h	0.17	investment rate of H (0.226)
Adjustment costs	$\rho_k = \rho_h$	0.10	symmetric
Productivity persistence, s.d.	ρ_s, σ_s	0.80, 0.10	Hennessy and Whited (2005)
Intangible financing capacity	θ	0.80	employee-equity share
Fixed operating cost	ξ	1.53	exit rate (0.050)
Intangible risk	σ_p	0.44	pre-2000 Intangible Gap (0.552)

Model Fit at the Benchmark

	Data: 1975–2000	Model benchmark
Intangible Gap	0.552	**0.557**
H/K, <15%	1.391	1.629
H/K, [15%,30%]	1.305	1.097
H/K, [30%,50%]	1.207	1.075
H/K, [50%,75%]	1.082	1.073
H/K, >75%	0.839	1.072
Median H/K	1.096	1.072
Investment rate of K	0.130	0.120
Investment rate of H	0.226	0.195
Exit rate	0.050	0.051
Employee equity to value	0.051	0.036

Explaining the Post-2000 Intangible Gap: Transition Dynamics



- Feed the measured rise in intangible risk: linearly over 1985–2000
- Sequence of unanticipated permanent shifts
- Gap is flat before 1990, keeps rising $\sim$a decade after risk flattens (slow cross-sectional adjustment via entry, exit, accumulation), plateaus $\sim$2010
- Model: 0.55 $\rightarrow$ **0.83**; data trend: 0.58 $\rightarrow$ 0.92 $\Rightarrow$ the bulk of the long-run rise

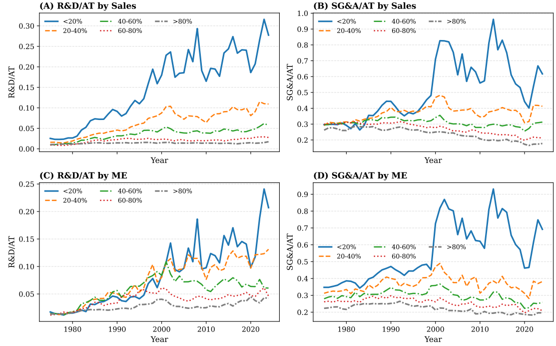
Financial-friction robustness

Conclusion

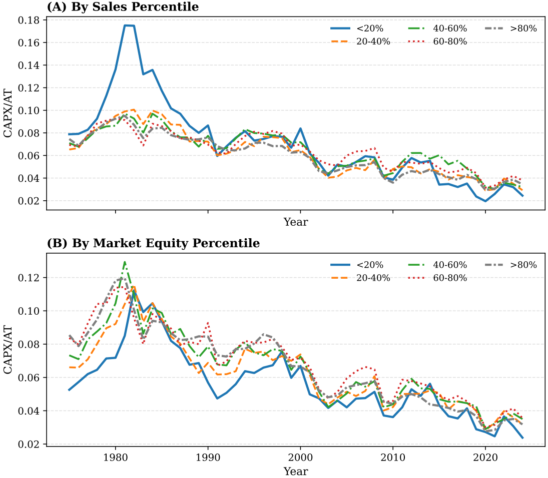
- Document a large and rising **intangible gap** between small and large U.S. firms: tripled since 1980, concentrated in H/K *volatility* more than in the mean
- Propose a mechanism that **differentiates** the effect of uncertainty between physical and intangible investment, via a multiplicative quality shock ω + endogenous exit option
- Combined with the intangible-financing channel ($d_t \geq -\theta h_t$), the model accounts for bulk of the post-2000 widening, with the rest unexplained by movements in θ

Appendix

Investment Rates by Firm Size



Intangible investment (XRD/AT, XSGA/AT) by size, 1975–2024.



Physical investment (CAPX/AT) by size, 1975–2024.

Small−large gap in R&D/AT: ≈0 in 1980 → 16 pp in 2024; in SG&A/AT: 17 → 50 pp. Physical investment rates are similar across size groups and declined secularly for all (7–12% → 2–4%). Back

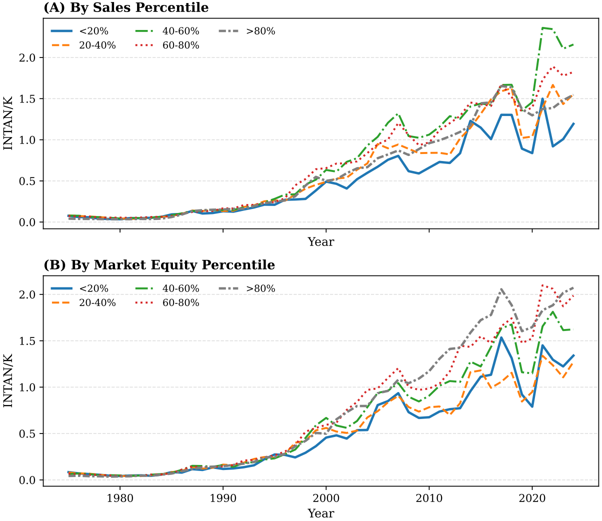
Within-Firm: H/K and Firm Size

$$\frac{H_{i,t}}{K_{i,t}} = \beta_0 + \beta_1 \log(size_{i,t}) + \alpha_i + \delta_t$$

	By Sales			By Market Equity		
	Total	R&D	SG&A	Total	R&D	SG&A
$\log(size)$	-1.091^{***}	-0.867^{***}	-0.379^{***}	-0.614^{***}	-0.574^{***}	-0.292^{***}
	(0.163)	(0.142)	(0.045)	(0.111)	(0.101)	(0.024)
Firm FE, Year FE	Yes					
Observations	170,090	170,426	170,548	168,018	168,338	168,460

Identified from a firm's *own* size changes: as a firm grows, its capital mix shifts toward physical capital. Robust to alternative denominators ($\log(H/K)$, $H/(H+K)$, H/AT, H per employee), industry$\times$year FE, age controls, excluding microcaps/recent IPOs, and IPO-cohort comparisons. Back

Purchased (Balance-Sheet) Intangibles: No Small-Firm Advantage



- INTAN/K rising for *all* size groups (Crouzet and Eberly, 2019)
- Small-firm advantage **absent**; gradient **reversed** by end of sample—largest quintile holds the highest ratio
- Consistent with large firms accumulating balance-sheet intangibles through **acquisitions**, while the intangible gap reflects *internally created* intangible capital

Back

Summary Statistics by Size Quintile (ME, 1975–2024)

Market equity	<20%	20–40%	40–60%	60–80%	>80%
H/K	4.46	3.81	3.31	2.65	2.06
H(SGA)/K	2.04	1.47	1.19	0.98	0.75
H(RD)/K	1.33	1.48	1.23	0.80	0.47
Stkco / ME (%)	1.29	0.83	0.50	0.33	0.20
Stkco / AT (%)	1.57	1.04	0.73	0.60	0.43
R&D / AT (%)	7.80	7.30	5.92	4.08	2.71
SGA / AT (%)	54.66	37.15	30.25	26.00	22.35
Age (found)	25.90	28.28	31.66	38.26	58.55
Tobin's q	1.36	1.55	1.66	1.76	1.89
ROA (%)	−29.79	−11.99	−3.69	1.88	5.79
Book Leverage (%)	28.14	23.73	22.61	23.34	24.22
Market Leverage (%)	29.36	23.99	22.21	21.31	19.62

Back

Testing the Mechanism: Measurement Details

Prediction: uncertainty raises *intangible*—not physical—investment, most strongly for firms *close to the exit margin*.

- **Ex-ante uncertainty exposure** (shift-share, predetermined):

$$U_{i,t} = \frac{\sum_c w^0_{i,c}\,\sigma_{c,t}}{\sum_c w^0_{i,c}}$$

  - $\sigma_{c,t}$: cross-patent std. of log patent value (Kogan et al., 2017) within CPC subclass c, 3-yr window; $w^0_{i,c}$: firm's patent mix over first five patenting years, fixed
  - Time variation from technology classes; cross-firm variation predetermined
  - Complementary: realized return volatility (Alfaro, Bloom, and Lin, 2024), 1992–2023

- **Proximity to the exit margin**, two proxies:
  - Standardized log market equity (small = near the margin)
  - Ex-ante **distress-exit score** $E_{i,t}$: logit of distress delisting (CRSP: liquidations, dropped) within 3 years on predetermined fundamentals; AUC = 0.83

Back

Uncertainty, Exit Risk, and the Composition of Investment

$$I^H_{i,t}/A_{i,t} = \beta\,(U_{i,t} \times \text{Exit}_{i,t}) + \gamma\, U_{i,t} + \lambda\, \text{Exit}_{i,t} + \alpha_i + \delta_t + \varepsilon_{i,t}$$

	Intangible inv. (R&D+SG&A)		Physical inv. (CAPX)	
$U\times$Exit	By size	By exit risk	By size	By exit risk
Shift-share, 1975–2024	-0.722^{***}	$+0.790^{***}$	-0.103^{***}	0.049
Realized vol., 1992–2023	-0.890^{***}	$+1.718^{***}$	$+0.094^{***}$	0.043^{*}

Size proxy: larger = *farther* from exit $\Rightarrow$ predicted $\beta < 0$; exit-risk proxy: larger = *closer* $\Rightarrow$ predicted $\beta > 0$. Firm & year FE; standardized regressors.

- 1 s.d. closer to the exit margin $\Rightarrow$ intangible investment +0.7–0.8 pp of assets per s.d. of uncertainty (shift-share); +0.9–1.7 pp (realized vol.)
- Level effect $\gamma \approx 0$ under shift-share $\Rightarrow$ effect operates *entirely* through the exit margin
- Physical investment: an order of magnitude smaller, unstable sign—the **differential** prediction

Full estimates Back

Mechanism Regressions: Full Estimates

	By Size (log ME)				By Exit Risk (E)			
	Total	R&D	SG&A	Phys.	Total	R&D	SG&A	Phys.
Panel A: Shift-Share Patent-Dispersion Exposure, 1975–2024								
Uncertainty × Exit	-0.722^{***}	-0.126^{***}	-0.584^{***}	-0.103^{***}	0.790^{***}	0.111^{**}	0.697^{***}	0.049
Uncertainty (U)	-0.061	-0.043	0.029	-0.042	-0.143	-0.052	-0.010	-0.052
Observations	100,128	100,277	100,254	99,337	82,842	83,066	82,867	82,078
Panel B: Realized Volatility, 1992–2023								
Uncertainty × Exit	-0.890^{***}	-0.163^{***}	-0.654^{***}	0.094^{***}	1.718^{***}	0.285^{***}	1.124^{***}	0.043^{*}
Uncertainty (U)	0.156	-0.051	0.068	-0.236^{***}	0.346^{*}	-0.138^{**}	0.663^{***}	-0.402^{***}
Observations	83,279	83,303	83,305	82,532	75,306	75,382	75,253	74,426
Firm FE, Year FE	Yes							

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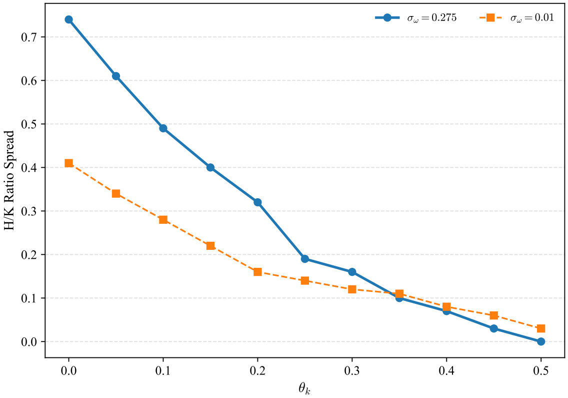
Robustness: Generalized Financial Frictions



H/K spread vs. θ_k, fixing $\theta_h = 0.8$. Blue: $\sigma_\omega = 0.275$; yellow: no quality shocks.

- **Extension:** $d_t \geq -(\theta_h h_t + \theta_k k_t)$—physical capital as collateral, on equity-like terms (a *generous* stress test)
- **Empirical range** from Lian and Ma (2021): $\theta_k \in [0.05, 0.10]$, an order of magnitude below $\theta_h = 0.8$ Details
- **Result:** in this range the model still delivers an H/K spread of ≈ 0.30–0.35—mechanism weakens only modestly
- Large θ_k: collateral substitutes, concavity (Jensen) loss dominates $\Rightarrow$ spread larger *without* volatility

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Calibration of θ_k – Details

Direct anchor (Lian–Ma Table 8):

$$\text{ABL/Assets} \approx 0.02$$

With $\text{Assets}/k \approx 2.4$ (since $\text{Assets} \approx k + h$, $h/k \approx 1.36$):

$$\theta_k \approx 0.02 \times 2.4 \approx 0.05$$

Pledgeability haircut:

- Advance rate $\approx$ 60% of book inventory
- Liquidation value $\approx$ 30–60% of book for fixed assets
- Effective max: $0.60 \times 0.50 = 0.30$

Plausible range: $\theta_k \in [0.05, 0.10]$. Both estimates are upper bounds (abstract from default/intermediation costs); the equity-equivalent θ_k is smaller still. Back

Microfoundation: Intangible Projects and Internal Diversification

- Each unit of intangible stock = one **project** (research line, prototype, training run, organizational initiative); $\tilde{h}$ = portfolio of $N = \tilde{h}$ projects
- Project quality ε_i i.i.d. with $\mathrm{Var}(\log \varepsilon_i) = \boldsymbol{\sigma_p^2}$; firm-wide factor u (same lab, technology bet, key personnel) with $\mathrm{Var}(\log u) = \sigma_c^2$
- Effective stock $\sum_i u\,\varepsilon_i = \omega\tilde{h}$, $\omega = u\,\bar{\varepsilon}_N$:

$$\sigma_\omega^2(\tilde{h}) \;\approx\; \sigma_c^2 \;+\; \frac{\sigma_p^2}{\tilde{h}}$$

- **Internal diversification**: large portfolios average away project draws $\rightarrow$ common-factor floor σ_c; small portfolios stay exposed
- Consistent with the size–volatility regularity (Stanley et al., 1996; Herskovic et al., 2016); granularity logic (Gabaix, 2011); Fact 2 confirms it for H/K itself

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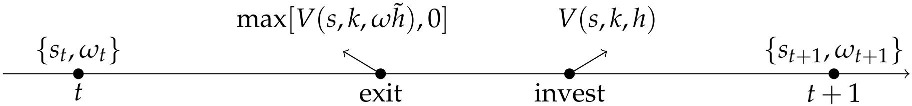
Within-Period Timing



Firm enters with $(s_t, k_t, \tilde{h}_t)$, observes shocks, decides exit, then invests $(k_{t+1}, \tilde{h}_{t+1})$.

Exit & Entry

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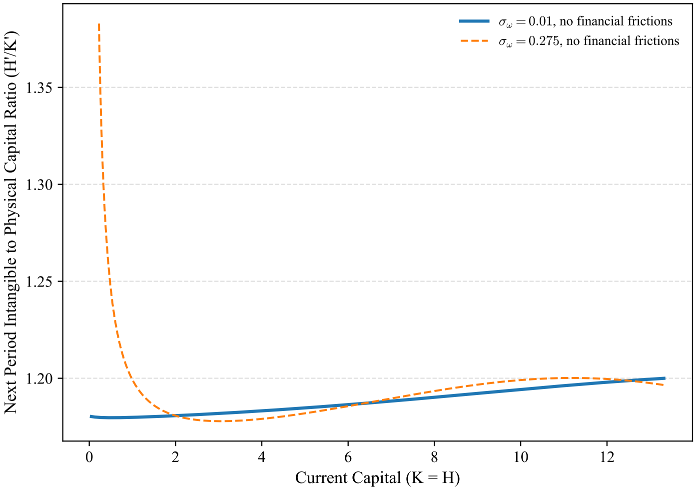
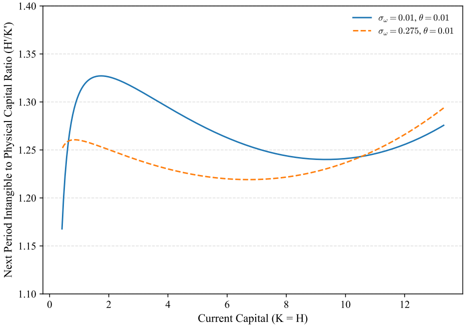
Policy Function: No Financial Frictions



H'/K' along the diagonal $K = H$; two levels of σ_ω (0.01 vs. 0.275), no financial frictions.

Policy Function: Severe Financing Frictions ($\theta = 0.01$)



H'/K' along $K = H$, $\sigma_\omega = 0.01$ vs. 0.275, with $\theta = 0.01$.

Exit and Entry: Details

Exit decision:

- Firm exits if $V(s_t, k_t, \omega_t \tilde{h}_t) < 0$
- Defines exit threshold $\omega^*(s_t, k_t, \tilde{h}_t)$:

$$\omega^* = \inf\{\omega : V(s, k, \omega\tilde{h}) \geq 0\}$$

- Firm bears fixed operating cost ξ each period it operates; ξ calibrated to match annual exit rate

Entry decision:

- Potential entrants draw s^0 from Γ, pay sunk cost c_e, and choose initial (k, h) and equity
- Free-entry condition $V^e(s^0) \geq c_e$ defines the entry threshold s_e^*
- Entry replaces exit: each exiting firm's slot is filled by an entrant; total mass of active firms normalized to one (Proposition 3: unique invariant distribution μ^*)

Back

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